MTH1065/1077 Group Assignment

Due: 23rd December, 2026

Groups

This assignment is to be completed in groups of 2–3 students. MTH1065 students and MTH1077 students can be in the same group. Each group should submit your group choice in Loop by 28th October (Wednesday) 11:59pm. Students who do not specify a group preference will be assigned to groups at random.

Submission

Please submit your attempts to the following tasks by 5:00 p.m. on Wednesday, 23 December 2026. The assignment must be submitted electronically to Loop. The submission should include a complete assignment coversheet (can be downloaded in Loop), your attempts in theoretical questions, and your outputs from your R programme in a single PDF and the R code for programming tasks.

Problem Setup

Under a risk–neutral measure, a stock price process $\{S_t : t \geq 0\}$ has dynamics given by a stochastic volatility model, i.e.

$$\begin{aligned} dS_t &= rS_t\,dt + \sqrt{V_t}S_t\,dW_t^S, && t > 0, \\ dV_t &= \kappa(\theta - V_t)\,dt + \xi(V_t)^{\frac{a}{4}}\,dW_t^V, && t > 0, \end{aligned}$$

with

$$a = \text{ your group number mod } 7 + 2.$$

where $\{V_t : t \geq 0\}$ is the volatility process, r is the risk–free rate, and W_t^S and W_t^V are Brownian motions which are correlated; denote their correlation coefficient by ρ.

Our goal is to price an option with S as the underlying given the following market model parameters:

S_0	V_0	r	ξ	θ	κ	ρ
110	0.30	0.04	0.50	0.06	1.80	-0.20

The option in question matures at time $T = 2$ and has a fixed strike price of $K = 110$. Each group has been given a conventional European put option with payoff $(K - S_T)_+$ where $(x)_+ = \max\{x, 0\}$ and one of the following **path–dependent** options to price:

$$\text{Let} \qquad A_T = \frac{1}{T}\int_0^T S_t\,dt, \quad m_T = \min_{0\leq t\leq T} S_t, \quad M_T = \max_{0\leq t\leq T} S_t.$$

- Fixed-strike Asian put: Payoff $= \max(K - A_T, 0)$
- Average-triggered put: Payoff $= (K - S_T)_+ \mathbf{1}_{\{S_T < A_T\}}$
- Down-and-out put: Payoff $= (K - S_T)_+ \mathbf{1}_{\{m_T > B_L\}}$
- Down-and-in put: Payoff $= (K - S_T)_+ \mathbf{1}_{\{m_T \leq B_L\}}$

- Up-and-out put: Payoff = $(K - S_T)_+ \mathbf{1}_{\{M_T < B_U\}}$
- Up-and-in put: Payoff = $(K - S_T)_+ \mathbf{1}_{\{M_T \geq B_U\}}$
- Double knock-out put: Payoff = $(K - S_T)_+ \mathbf{1}_{\{m_T > B_L,\ M_T < B_U\}}$
- Fixed-strike lookback put: Payoff = $(K - m_T)_+$

where the barrier levels are $B_L = 80$ and $B_U = 130$.

Tasks

(a.) (i.) Write a function **SVEM** which takes as input M (number of paths) and $N = T/h$ (number of steps, where h is the step size), and outputs an $(N+1) \times M$ matrix in which each column is a simulated path of the stock price process $\{S_t : t \geq 0\}$ in the market described above using the Euler–Maruyama scheme. Explain and justify any corrections or adjustments you make to the standard Euler–Maruyama scheme.

(ii.) Write a function **SVMil** which takes as input M (as above) and N (as above), and outputs an $(N+1) \times M$ matrix in which each column is a simulated path of the stock price process $\{S_t : t \geq 0\}$ in the market described above using the Milstein scheme. Explain and justify any corrections or adjustments you make to the standard Milstein scheme.

(b.) (i.) Use both of the functions you have written in part (a.) to price your conventional European put (i.e. payoff is $(K - S_T)_+$) for various values of M and N (small to large) using the Monte Carlo Method. Tabulate your results neatly and briefly comment on them.

(ii.) Assume that the stock follows the Black-Scholes model with $\sqrt{V_t} = \sqrt{\theta}$. Calculate the price of the European put using the Black-Scholes formula. Briefly comment on the difference between this price and the price in (b.) part(i.).

(c.) (i.) Use both of the functions you have written in part (a.) to price your given path-dependent option for $M = 10000$ and $N = 1000$ using the Monte Carlo method. Tabulate your results neatly and briefly comment on them. Briefly comment on the difference of the price of your given option from the price of conventional put in part (b.)(i.).

(ii.) Calculate the standard error and 95% (asymptotic) confidence interval for the option prices you have calculated in (c.) part (i.).

(iii.) Explore how increasing or decreasing N (with M fixed) impacts the price of your path-dependent option. Discuss your findings.

(d.) In this part, we will apply control variate method with the control S_T to price your given path-dependent option

(i.) Use both of the functions you have written in part (a.) to simulate $M_0 = 5000$ paths with $N = 1000$ as a pilot simulation to estimate the control coefficient.

(ii.) Using a new independent simulation with (M=10,000) and (N=1,000) from both Euler and Milstein schemes respectively to estimate the path-dependent option price using the control-variate method and the coefficient obtained in Part (d.)(i.). Report the control-variate estimates, their standard errors and their 95% asymptotic confidence intervals. Calculate the estimated variance-reduction factor and comment on the effectiveness of the control variable.

(iii.) Repeat parts (d.)(i.) and (d.)(ii.) by combining the control-variate and antithetic-variate methods. Use half number of independently generated paths $M/2$.